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HCM·UTE

LECTURE 7 • ANALOG CMOS IC DESIGN

Noise Analysis Fundamentals

Noise in time & frequency domains • Resistor and MOSFET noise • SNR and input-referred noise • CS amplifier example

Source: H. A. Omran, "Analog IC Design — Lecture 17: Noise Analysis Fundamentals," Ain Shams University, 2025

AGENDA

Outline of This Lecture

Five technical parts, from noise statistics to a complete amplifier noise analysis

I **Noise in Time and Frequency Domains**

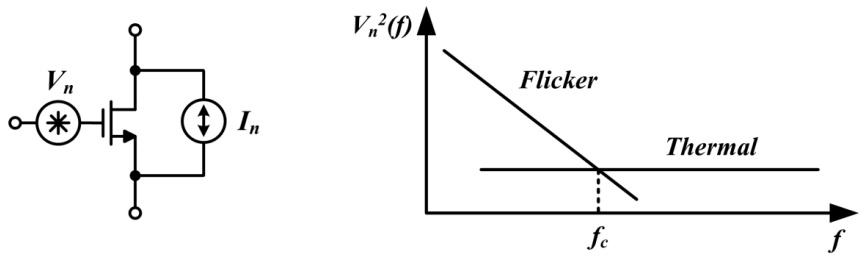
II **Resistor Thermal Noise and kT/C Noise**

III **MOSFET Thermal and Flicker Noise**

IV **SNR and Input-Referred Noise**

V **Noise Analysis Example: CS Amplifier**

VI **Conclusion**



Flicker vs. thermal noise PSD and the corner frequency f_c

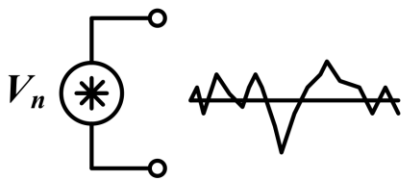
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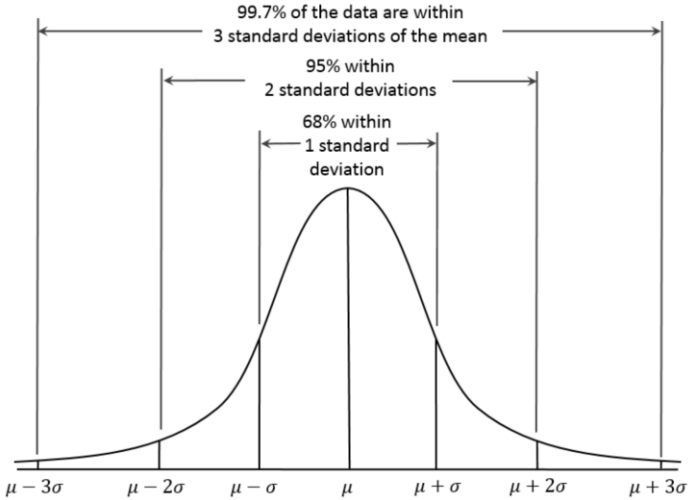
Noise in Time and Frequency Domains

- Statistical view: Gaussian, zero mean, σ = rms value
- Noise power, mean-square and rms noise voltage
- Power spectral density (PSD), white noise and noise shaping
- Types of noise and internal noise mechanisms

Noise in the Time Domain: a Statistical View

- Noise is an unwanted random signal: its instantaneous value cannot be predicted
- Its statistical distribution can be modeled — normal (Gaussian)
- Mean is zero: the average noise voltage in time is 0
- Variance σ^2 = mean-square value; standard deviation σ = rms value
- Peak-to-peak instantaneous noise is usually within $\pm 3\sigma$ (99.7 %)





[Omran, Lec. 17, slides 16–17]

Noise Power and RMS Noise Voltage

- Noise is non-periodic: its average power is defined by a time average over $T \to \infty$
- The mean-square noise voltage is proportional to the average noise power
- The rms noise voltage equals the standard deviation σn of the noise distribution
- Noise is therefore specified by its mean-square (V^2) or rms (V) value, never by its instantaneous value

$$P_{avg} = \frac{1}{R_L} \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} V_n^2(t)\, dt = \frac{\overline{V_n^2}}{R_L}$$

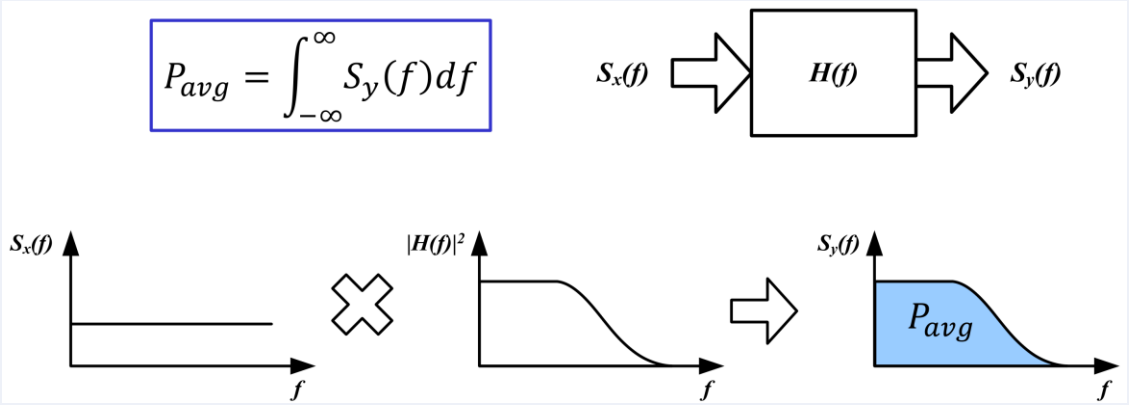
$$\overline{V_n^2} = \sigma_n^2 \propto P_{avg}, \qquad V_{n,rms} = \sqrt{\overline{V_n^2}} = \sigma_n$$

[Omran, Lec. 17, slide 18]

Noise in the Frequency Domain and Noise Shaping

- **PSD $S_x(f)$:** average power (mean-square value) in a 1-Hz band around f [W/Hz or V^2/Hz]; voltage noise density $V_n(f) = \sqrt{S_x(f)}$ [V/$\sqrt{Hz}$]
- **White noise:** the PSD has the same value at all frequencies
- **Noise shaping:** the spectrum is shaped by the system, $S_y(f) = S_x(f) \cdot |H(f)|^2$; output noise power = area under $S_y(f)$
- **Wide BW is not good BW:** a wider bandwidth integrates more noise — the amplifier BW should just fit the signal

$$P_{avg} = \int_{-\infty}^{\infty} S_y(f)\,df$$



[Omran, Lec. 17, slides 19–21] • *[Razavi, 2017]*

Types of Noise and Internal Noise Mechanisms

1

External (Interference) Noise

Man-made: EM interference, power-supply noise. Can be eliminated by careful design, layout and shielding.

2

Thermal (Johnson / Nyquist) Noise

Thermal excitation of carriers; white PSD; independent of DC current; in all resistive elements.

3

Shot Noise

Non-smooth DC current (flow of individual carriers); white PSD; occurs in pn-junctions (hence BJTs).

4

Flicker (1/f) Noise

Traps in the semiconductor affecting DC current flow; a significant noise source in MOSFETs.

Internal noise (thermal, shot, flicker) is inherent to the devices: it can be reduced but not eliminated • [Omran, Lec. 17, slides 22–23]

II

Resistor Thermal Noise and kT/C Noise

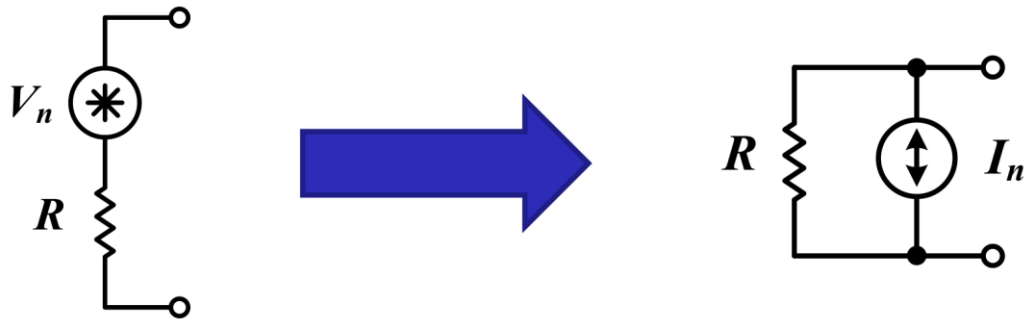
- Thevenin and Norton noise models: 4kTR and 4kT/R
- Noise in an RC circuit: the kT/C result
- Equivalent noise bandwidth B_N

Resistor Thermal Noise: Thevenin and Norton Models

- From thermodynamics, the thermal-noise PSD of a resistor is white
- $k = 1.38\times10^{-23}$ J/K; $kT = 4.14\times10^{-21}$ J at T = 300 K
- **R = 1 kΩ:** $V_n(f) \approx 4$ nV/√Hz and $I_n(f) \approx 4$ pA/√Hz
- **Scaling:** $V_n(f) \approx 4$ nV/√Hz × √(R / 1 kΩ); $I_n(f) \approx 4$ pA/√Hz × √(1 kΩ / R)
- The Thevenin model (series V_n) can be converted to the Norton model (parallel I_n)

$$\overline{V_n^2}(f) = 4kTR$$

$$\overline{I_n^2}(f) = \frac{\overline{V_n^2}(f)}{R^2} = \frac{4kT}{R}$$

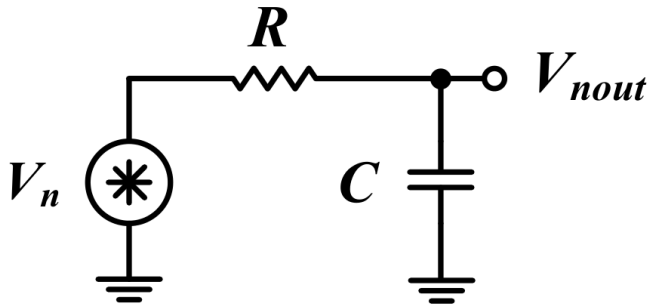


[Omran, Lec. 17, slides 25–26]

Noise in an RC Circuit: the kT/C Result

Resistors never exist alone — the bandwidth is always limited by a capacitor

$$\overline{V_{n,out}^2} = \int_0^\infty 4kTR \left| \frac{1}{1+j2\pi fRC} \right|^2 df = \frac{kT}{C}$$



Independent of R

↑R: higher PSD 4kTR, but lower BW

C = 1 pF → $V_{n,rms}$ ≈ 64 µVrms

$V_{n,rms} \approx 64\ \mu Vrms \times \sqrt{(1\ pF / C)}$

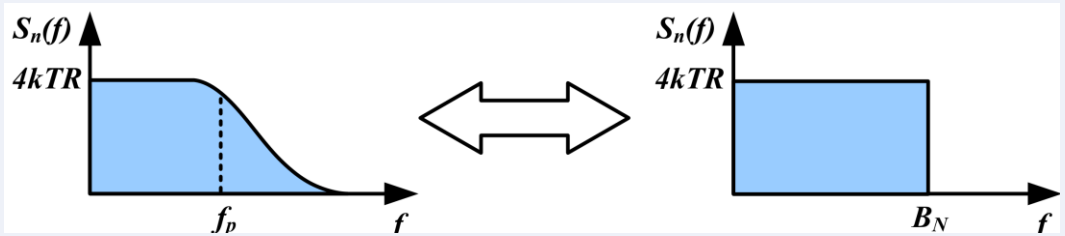
- The total (integrated) rms noise depends only on the capacitance: kT/C
- Only a larger capacitance lowers the rms noise of an RC network

Equivalent Noise Bandwidth B_N

- **Definition:** B_N is the width of a brick-wall response whose area equals the area under the actual PSD
- Total rms noise = (white PSD) × B_N — a convenient shortcut for any filter shape
- **First-order (RC) system:** $B_N = (\pi/2)\cdot f_p \approx 1.57\ f_p$, i.e., larger than the −3 dB bandwidth f_p

$$\overline{V_{n,rms}^2} = \int_0^\infty S_n(f)\,df = 4kTR \cdot B_N = \frac{kT}{C}$$

$$B_N = \frac{1}{4RC} = \frac{\pi}{2} f_p$$



[Omran, Lec. 17, slide 29]

III

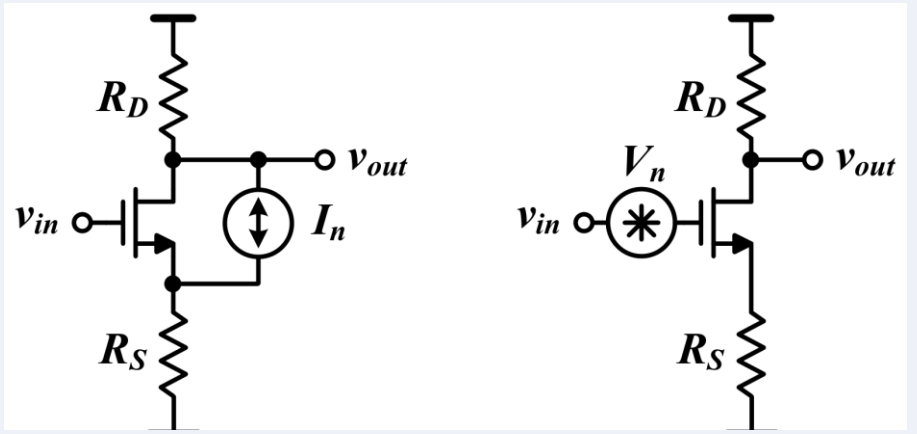
MOSFET Thermal and Flicker Noise

- Channel thermal noise: $4kT\gamma g_m$
- Referring the noise current to the gate
- Flicker (1/f) noise and the corner frequency f_c
- Summary of noise models

MOSFET Channel Thermal Noise and Its Gate-Referred Model

- The resistive channel generates thermal noise — like a resistor with $R = 1/(\gamma g_m)$
- **γ:** thermal-noise coefficient; $\gamma \approx 2/3$ for long-channel devices, close to 1 for short-channel devices
- The drain noise current can be referred to the gate through g_m (not G_m): same short-circuit output current in both models
- Valid at zero gate current, i.e., at low/medium frequencies

$$\overline{I_n^2}(f) = 4kT\gamma g_m \quad \Longleftrightarrow \quad \overline{V_n^2}(f) = \frac{4kT\gamma}{g_m}$$



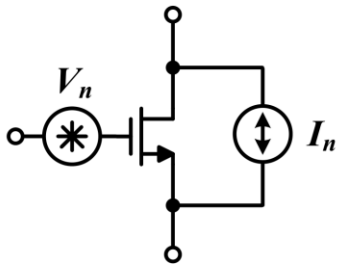
[Omran, Lec. 17, slides 31–32]

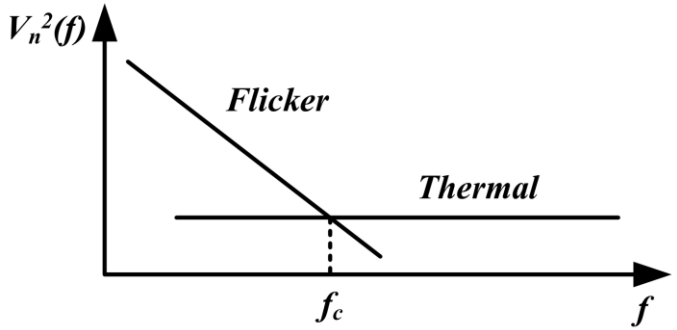
MOSFET Flicker (1/f) Noise and Corner Frequency

- Mainly due to dangling bonds at the oxide/silicon interface
- **K:** flicker-noise coefficient; PSD falls at −10 dB/dec
- Reduced by increasing the device area WL
- PMOS usually has much less flicker noise than NMOS
- **Corner f_c:** frequency where flicker and thermal PSDs are equal — tells which noise dominates a given signal band
- f_c can reach 100s of MHz in deep-submicron nodes

$$\overline{V_n^2}(f) = \frac{K}{C_{ox}WL} \cdot \frac{1}{f}$$

$$4kT\gamma g_m = \frac{K}{C_{ox}WL} \cdot \frac{g_m^2}{f_c} \Rightarrow f_c = \frac{K}{C_{ox}WL} \cdot \frac{g_m}{4kT\gamma}$$





[Omran, Lec. 17, slides 33–34]

Noise Models Summary

Noise source	Model (spectral density)	Key numbers / remarks
Resistor thermal	$V_n^2(f) = 4kTR$ (series) $I_n^2(f) = 4kT/R$ (parallel)	$V_n \approx 4\ nV/\sqrt{Hz} \times \sqrt{(R/1\ k\Omega)}$ $I_n \approx 4\ pA/\sqrt{Hz} \times \sqrt{(1\ k\Omega/R)}$
MOSFET thermal	$I_n^2(f) = 4kT\gamma g_m$ (drain) $V_n^2(f) = 4kT\gamma/g_m$ (gate)	$\gamma \approx 2/3$ (long channel), ≈ 1 (short channel)
MOSFET flicker	$V_n^2(f) = K / (C_{ox}WL \cdot f)$ (gate)	−10 dB/dec; ↓ with larger WL; PMOS < NMOS
RC network (rms)	$V_{n,rms}^2 = 4kTR \cdot B_N = kT/C,\ B_N = 1/(4RC)$	$V_{n,rms} \approx 64\ \mu Vrms \times \sqrt{(1\ pF/C)}$

[Omran, Lec. 17, slide 35]

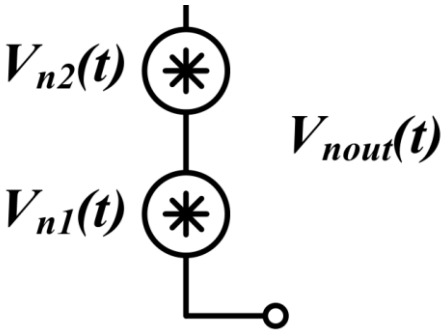


SNR and Input-Referred Noise

- Signal-to-noise ratio (SNR) and its dB form
- Adding uncorrelated noise sources
- Output-referred vs. input-referred noise

Signal-to-Noise Ratio and Multiple Noise Sources

- **SNR:** ratio of signal power to noise power, usually expressed in dB
- **Example:** $V_{sig,rms}$ = 100 mVrms, $V_{n,rms}$ = 100 µVrms → SNR = 60 dB
- Noise voltages add in the time domain, but rms values cannot be added: $V_{n,out,rms} \neq V_{n1,rms} + V_{n2,rms}$
- Uncorrelated sources add in power; the largest contributor dominates ($3^2 + 1^2 \approx 3^2$)



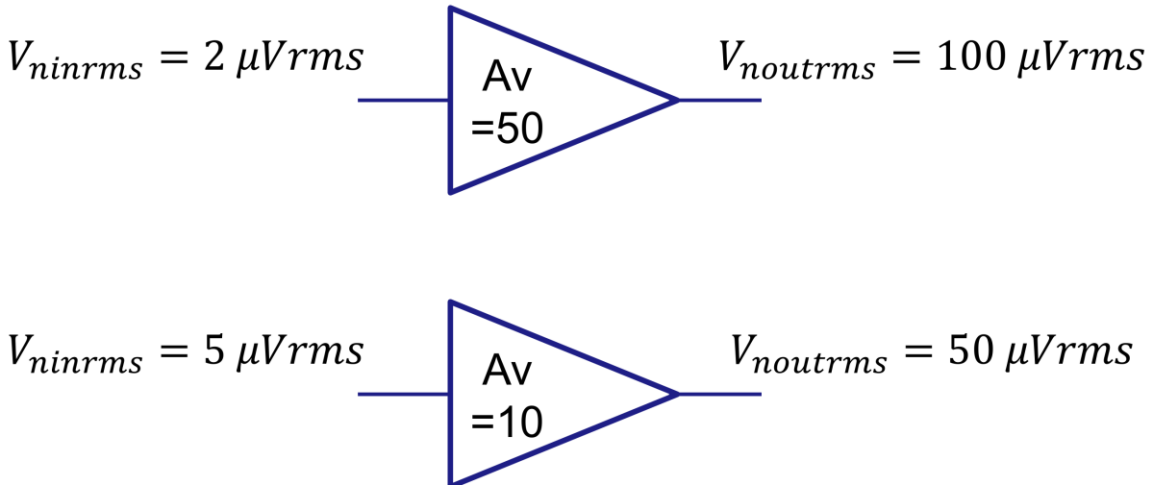
$$SNR = \frac{P_{signal}}{P_{noise}} = \frac{V_{sig,rms}^2}{V_{n,rms}^2}, \qquad SNR_{dB} = 10\log\frac{P_{signal}}{P_{noise}} = 20\log\frac{V_{sig,rms}}{V_{n,rms}}$$

$$V_{n,out}(t) = V_{n1}(t) + V_{n2}(t) \Rightarrow V_{n,out,rms}^2 = V_{n1,rms}^2 + V_{n2,rms}^2 \quad \text{(uncorrelated)}$$

[Omran, Lec. 17, slides 37–38]

Output-Referred vs. Input-Referred Noise

- Which amplifier has lower noise? Output-referred noise does not allow a fair comparison
- Output noise depends on the amplifier gain — but the signal is multiplied by the gain as well
- **Fair comparison:** refer the noise to the input, $V_{n,in,rms} = V_{n,out,rms} / |A_v|$
- **Result:** the $A_v = 50$ amplifier (2 µVrms) is quieter than the $A_v = 10$ one (5 µVrms), despite its larger output noise



[Omran, Lec. 17, slides 39–40]

V

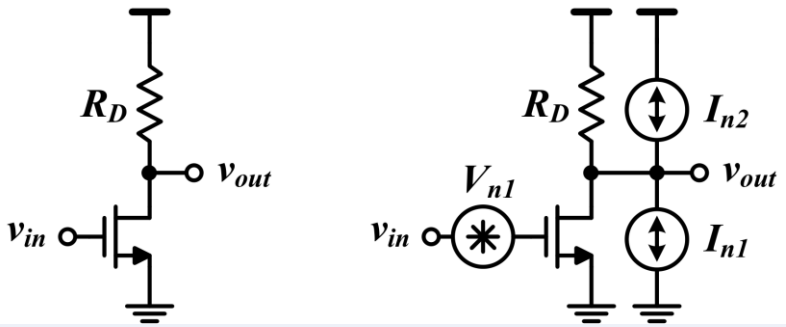
Noise Analysis Example: CS Amplifier

- Output and input-referred noise density of a CS stage
- A systematic noise-analysis procedure
- RMS output noise, $kT\theta/C$ and SNR
- Noise–power and noise–speed trade-offs

CS Amplifier with Resistive Load: Output Noise Density

- Deactivate the input signal and identify the noise sources
  - R_D: thermal noise ($I_{n2}^2 = 4kT/R_D$)
  - M1: thermal noise ($I_{n1}^2 = 4kT\gamma g_m$) + flicker noise (V_{n1}^2 at the gate)
- Apply superposition (noise is a small signal) and sum the output noise powers (r_o neglected)

$$\overline{V_{n,out}^2}(f) \approx \left(4kT\gamma g_m + \frac{K}{C_{ox}WL} \cdot \frac{g_m^2}{f} + \frac{4kT}{R_D}\right) R_D^2$$



[Omran, Lec. 17, slides 42–44]

Input-Referred Noise Density of the CS Stage

- For a fair comparison, divide the output density by $A_v^2 = (g_m R_D)^2$
- **Thermal term $4kT\gamma/g_m$:** lowered only by larger g_m, i.e., more bias current — a fundamental power–noise trade-off
- **Flicker term:** set by device area WL — independent of bias
- **Load term $4kT/(g_m^2 R_D)$:** suppressed by a large gain $g_m R_D$

$$\overline{V_{n,in}^2}(f) = \frac{\overline{V_{n,out}^2}(f)}{A_v^2} = \frac{4kT\gamma}{g_m} + \frac{K}{C_{ox}WL} \cdot \frac{1}{f} + \frac{4kT}{g_m^2 R_D}$$

[Omran, Lec. 17, slide 45]

Noise Analysis Procedure

Step	Action
1. Deactivate	Deactivate (null) the input signal
2. Identify	Identify the dominant noise sources and model each as $V_n^2(f)$ or $I_n^2(f)$
3. Transfer	Find the output noise density of each source: $V_{n,out,x}^2(f)$
4. Integrate	RMS output noise of each source: $V_{n,out,rms,x}^2 = V_{n,out,x}^2(f) \times B_{N,x}$
5. Sum	Total rms output noise: $V_{n,out,rms,tot}^2 = V_{n,rms,1}^2 + V_{n,rms,2}^2 + \ldots$
6. Refer to input	$V_{n,in,rms,tot}^2 = V_{n,out,rms,tot}^2 / A_v^2$ (for low Z_{in}, also add the input-referred noise current)

[Omran, Lec. 17, slide 46]

RMS Output Noise and SNR of the CS Stage

- Assume thermal noise dominates and the BW is limited by C_L ($B_N = 1/(4R_DC_L)$)
- **General form:** rms output noise = kTθ/C, with C the BW-limiting capacitor and θ > 1 topology-dependent
- **SNR:** sinusoidal input of amplitude V_p; $V_{RD} = V_{DD}/2$ maximizes the output swing; $g_mR_D = 2V_{RD}/V^*$
- Higher V_{DD} improves SNR — but V_{DD} has scaled from >10 V to sub-1 V, making low-voltage design challenging

$$V_{n,out,rms}^2 \approx 4kTg_m\left(\gamma+\frac{1}{g_mR_D}\right)R_D^2\cdot\frac{1}{4R_DC_L}=\frac{kT\theta}{C_L},\qquad \theta=1+\gamma g_mR_D=1+\gamma|A_v|$$

$$SNR\approx\left(\frac{V_p}{\sqrt{2}}g_mR_D\right)^2\frac{C_L}{kT\theta}=\frac{2V_p^2V_{RD}^2C_L}{V^{*2}kT\theta}$$

$$V_{RD}=\frac{V_{DD}}{2}\quad\Rightarrow\quad SNR\approx\frac{V_p^2V_{DD}^2C_L}{2V^{*2}kT\theta}$$

[Omran, Lec. 17, slides 47–49]

Noise–Power and Noise–Speed Trade-offs

$$SNR \approx \frac{V_p^2 V_{DD}^2 C_L}{2V^{*2} kT\theta}, \qquad GBW = \frac{g_m}{C_L}$$

To gain +6 dB SNR	Speed spec fixed (noise–power)	Power spec fixed (noise–speed)
Load capacitance C_L	× 4	× 4
Transconductance g_m	× 4 (to keep GBW = g_m/C_L)	unchanged
Bandwidth GBW	unchanged	÷ 4
Power dissipation	× 4 (assuming constant V*)	unchanged
Other knob	—	Lower V* raises g_m at the same current, but f_T drops → lower speed

+6 dB SNR ≈ 1 bit of resolution, in a system limited by thermal noise • [Omran, Lec. 17, slides 50–51]

VI. CONCLUSION

Key Takeaways

1. Noise is random: it is described statistically (Gaussian, zero mean, σ = rms) and spectrally (PSD); uncorrelated sources add in power

2. Resistor thermal noise 4kTR is white; the total rms noise of an RC circuit is kT/C, independent of R

3. MOSFET noise = channel thermal noise $4kT\gamma g_m$ + flicker noise $K/(C_{ox}WL \cdot f)$; the corner f_c tells which one dominates

4. Compare circuits by input-referred noise; for the CS stage $V_{n,in}^2 \approx 4kT\gamma/g_m + K/(C_{ox}WL \cdot f) + 4kT/(g_m^2 R_D)$

5. Thermal-noise-limited SNR $\propto V_{DD}^2 C_L/(kT\theta)$: +6 dB SNR costs 4× C_L and either 4× power or 4× lower GBW